Time: 40 minutes

Exercise: The "Eco-Fleet" Carbon Tracker

Goal: Build a system that calculates the environmental impact of different vehicles using a shared interface.

The Scenario

A local delivery company, **GreenLogistics**, wants to track the carbon footprint of their fleet. Because their fleet includes gas trucks, electric vans, and hybrids, the math for calculating emissions is different for every vehicle. However, the Fleet Manager needs a single report that shows the footprint for *every* vehicle regardless of its type.

Task 1: The Blueprint (10 Minutes)

Create an **Abstract Base Class (ABC)** named Vehicle.

- **Attributes:** Every vehicle must have a `model_name (string)` and a `kilometers_driven (float)`.
- **Abstract Method:** Define an abstract `method @abstractmethod` called `calculate_emissions(self)`. It should return a float.

Task 2: The Implementation (15 Minutes)

Create two subclasses that inherit from Vehicle. You must use `super().__init__()` to handle the shared attributes.

1. **GasTruck**:
    - **New Attribute:** `fuel_efficiency` (represented as km per liter).
    - **Override** `calculate_emissions()`: The formula is:

$$\text{Emissions} = (\text{kilometers_driven}/\text{fuel_efficiency}) \times 2.31$$

2. **ElectricVehicle**:
    - **New Attribute:** `energy efficiency` (kWh per km).
    - **Override** `calculate_emissions()`**:** Even though EVs have no tailpipe emissions, the company adds a "grid tax" for the electricity used. The formula is:

$$\text{Emissions} = \text{kilometers_driven} \times \text{energy_efficiency} \times 0.45$$

Task 3: Polymorphism & Reporting (15 Minutes)

Now, act as the "User" of these classes.

1. Create a list called `delivery_fleet` containing:
    - One `GasTruck` (e.g., "Ford F-150", 500km driven, 8km/L efficiency).
    - One `ElectricVehicle` (e.g., "Tesla Semi", 500km driven, 1.2 kWh/km efficiency).
2. Write a function called `generate_environmental_report(vehicles: list[Vehicle])`.
3. Inside the function, loop through the list. For each vehicle, print:

`"Vehicle: [Model Name] | Total Emissions: [Result] kg CO2"`

4. **Crucial Check:** Your function should only call `calculate_emissions()`